

ReProspect - A framework for reproducible prospecting of CUDA applications

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Summary

ReProspect is a Python framework for prospecting CUDA code, designed to ensure reproducibility through a fully programmatic approach. Prospecting encompasses three complementary ways of characterizing CUDA-based libraries and software components: how they interact with the CUDA runtime through API tracing, how kernels perform through kernel profiling, and how source constructs translate into machine code through binary analysis.

ReProspect builds on NVIDIA tools: Nsight Systems, Nsight Compute, and the CUDA binary utilities. It streamlines data collection and extraction using these tools, and it complements them with new functionalities for programmatic analysis of these data, thus making it possible to encapsulate the entire prospecting analysis in a single Python script.

Statement of need

HPC software development strives to achieve performance and sustain it over time as hardware and software continue to evolve. Yet the modern programming landscape relies on complex software stacks and compiler toolchains that make it increasingly difficult to reason about code behavior. Developers therefore need tools that support continuous assessment of their code and help them unravel the implications of their design decisions and changes.

Beyond *ad hoc* investigations, fully programmatic tools allow analyses to be integrated across the development cycle and open a range of new use cases. For instance, they support collaborative development by enabling developers to share concise, reproducible analyses that motivate design decisions and help reviewers grasp the impact of proposed changes. They also enable new types of tests in CI/CD pipelines that go beyond traditional output-correctness validation — such as confirming expected API call sequences for key library functionalities, verifying memory traffic for custom data layouts via kernel profiling, or validating the presence of specific instruction patterns in machine code. Finally, they can act as a framework for structuring research artifacts and documenting analyses, enabling others to reproduce and build upon prior work more effectively.

For the CUDA stack, NVIDIA provides a set of proprietary tools guaranteed to be up-to-date with their software and hardware. The runtime analysis tools Nsight Systems (NVIDIA Corporation, 2025d) and Nsight Compute (NVIDIA Corporation, 2025c) are designed for API tracing and kernel profiling, respectively. They both provide a GUI for exploring the results, as well as a low-level Python API for accessing the raw data. The CUDA binary utilities (NVIDIA Corporation, 2025a) provide command-line access to machine code (SASS or PTX (NVIDIA Corporation, 2025f)) and other information embedded in the binaries. However, while these tools allow raw data to be extracted, they do not themselves provide the infrastructure for effective programmatic analysis.

40 There is thus a need for a framework that builds on the NVIDIA tools and augments them to
41 enable fully programmatic analysis. ReProspect addresses this need and is designed to support
42 the aforementioned new use cases. It allows developers to write well-structured, concise Python
43 scripts focused on analysis results. It structures the collected data to be readily amenable
44 to test assertions, and introduces new capabilities such as matchers for instruction sequence
45 patterns in machine code extracted from binaries.

46 State of the field

47 Several well-established open-source tools are already available. Caliper (Boehme et al., 2016)
48 can intercept CUDA API calls through the NVIDIA CUPTI library (NVIDIA Corporation,
49 2025b). It can interface with the Python package Hatchet (Bhatele et al., 2019) to organize
50 results into a hierarchical data structure. Thicket (Brink et al., 2023) adds kernel profiling
51 support through Nsight Compute, with a primary focus on exploratory data analysis of multi-run
52 performance experiments. HPCToolkit (Zhou et al., 2021) is another comprehensive suite
53 designed for large-scale parallel systems. It includes CUDA API tracing through CUPTI and
54 kernel profiling through PAPI (Terpstra et al., 2010), and it has binary analysis capabilities to
55 attribute performance data to calling contexts. It has a visual interface, and it can output
56 raw performance data for programmatic analysis, e.g. using Hatchet. Score-P (Knüpfer et
57 al., 2012) integrates multiple performance analysis tools in a common infrastructure. It can
58 record CUDA API calls and GPU activities through CUPTI and provides standardized data
59 formats. By contrast, ReProspect focuses on concise programmatic analysis of individual units
60 of functionality based on the outputs of NVIDIA tools.

61 Several tools that operate on SASS embedded in CUDA binaries already exist, each with a
62 distinct focus. CuAssembler (Cloudcores, 2023) and CuAsmRL (He & Yoneki, 2025) focus on
63 modifying the SASS code. NVBit (Villa et al., 2019) is a dynamic instrumentation library for
64 CUDA binaries. However, none of these tools provides a SASS matching framework.

65 Miasm (Desclaux, 2012) is a Python reverse-engineering framework for CPU assembly: it
66 can lift instructions into an intermediate representation (IR) to support higher-level analysis
67 and binary modification. The Miasm expressions are conceptually close to the ReProspect
68 matching framework. However, CPU assembly differs significantly from SASS, which depends
69 on the architecture generation and embeds more information, such as scheduling control codes,
70 thread predicates, and memory space specifiers.

71 Software design

72 ReProspect is organized into three main components: API tracing, kernel profiling, and binary
73 analysis (Figure 1).

74 Each component is designed to allow the entire analysis to be encapsulated into a concise
75 Python script. This includes launching the underlying analysis tool, collecting the output into
76 Python data structures, and performing the subsequent analysis.

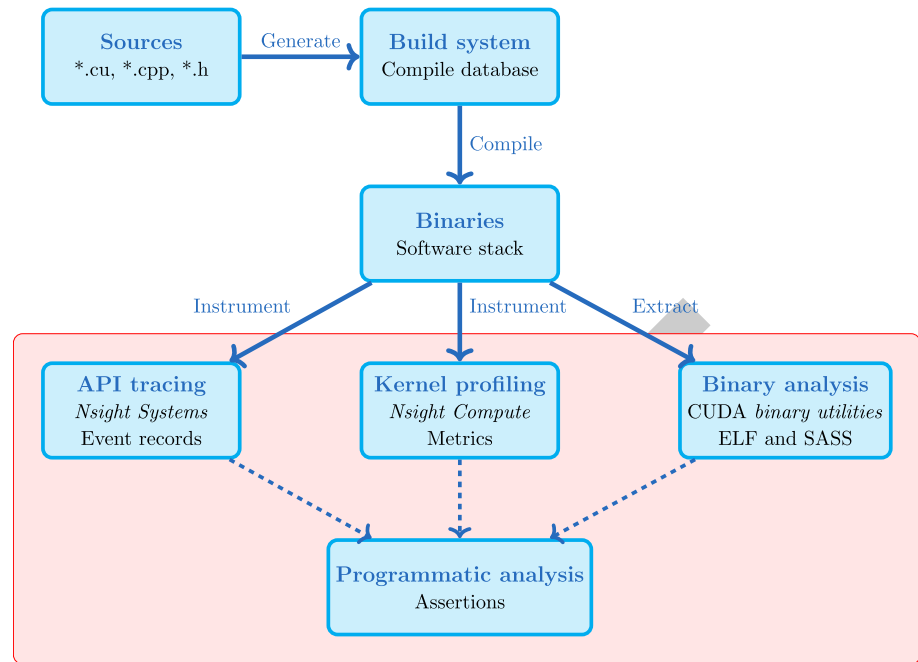


Figure 1: Overview of ReProspect.

API tracing and kernel profiling

The ReProspect Command and Session classes streamline launching Nsight Systems and Nsight Compute to collect a focused set of metrics most relevant for the analysis. The collected data are gathered in a Report, queryable by NVTX range annotations (NVIDIA Corporation, 2025e), readily amenable to test assertions.

To avoid unnecessary re-runs, ReProspect provides a Cacher that can serve the Report from a database.

The following snippet illustrates how the design allows a kernel profiling analysis to be scripted concisely, up to a test assertion—here, that the global stores of a kernel that fills a buffer with values of type char are fully coalesced:

```

metrics = (*L1TEXCache.GlobalStore.Instructions.create(),
           *L1TEXCache.GlobalStore.Requests.create(),
           *L1TEXCache.GlobalStore.Sectors.create())

nc = Session(command=Command(executable=executable, metrics=metrics,
                             output=workdir / executable.name,
                             nvtx_includes=('domain@start_end_range',)))
nc.run(cwd=workdir)

results = Report(command=nc.command).extract_results_in_range(metrics=metrics)
_, m = results.query_single_next_metrics(accessors=('kernel_range',))
assert m['L1/TEX cache global store requests'] \
       == m['L1/TEX cache global store sass instructions']
assert m['L1/TEX cache global store sectors'] \
       == m['L1/TEX cache global store requests']
  
```

The same design serves API tracing: the corresponding Report retrieves event records from the relational database generated by Nsight Systems, with accessors for selecting event records by NVTX range and for correlating them with data across database tables.

90 Binary analysis

91 ReProspect provides a set of tools for extracting and analysing the content of CUDA binaries.
92 The CuObjDump and NVDisasm classes drive and parse the output of the underlying CUDA
93 binary utilities to retrieve the SASS code and resource usage of kernels (e.g. registers, constant
94 memory).

95 The ELF class decodes ELF-formatted sections to extract complementary information, including
96 the symbol table, toolchain metadata (from the `.note.nv.tkinfo` section), and kernel attributes
97 such as launch bounds (from the `.nv.info.<kernel>` section) (Hayes et al., 2019).

98 Beyond data extraction, ReProspect provides an extensible framework for inspecting machine
99 code for expected instruction patterns. This framework is structured as a hierarchy of matchers.
100 At the lowest levels, matchers analyse instructions and their components (opcodes, modifiers,
101 and operands). These matchers can be composed into instruction sequence matchers, enabling
102 the identification of more intricate patterns.

103 One of the key challenges with SASS matching is the evolution of the CUDA instruction set and
104 compiler toolchains. ReProspect addresses this challenge by abstracting away architecture- and
105 toolchain-specific details at each level of the matcher hierarchy. For example, AddressMatcher
106 matches a memory address operand, adjusting the expected address format for the target
107 architecture. Then, at the instruction level, LoadMatcher matches loads from memory; it
108 uses AddressMatcher for the memory address and thus needs only to adjust the opcode and
109 modifiers for the target architecture. By extending this hierarchical design to instruction
110 sequence and basic block matchers, ReProspect enables robust, composable matching across
111 CUDA architectures and compiler toolchains.

112 The following snippet illustrates how matchers can be composed to assert the presence or
113 absence of a 16-bit floating-point code path, e.g. in the SASS codes in Table 1:

```
arch = NVIDIAArch(...)
instructions = Decoder(...)
cfg = ControlFlow.analyze(instructions)

matcher_ldg = instructions_contain(instructions_are(
    LoadGlobalMatcher(arch, size=16, extend='U', readonly=False),
    LoadGlobalMatcher(arch, size=16, extend='U', readonly=False),
))
blk, matched_ldg = BasicBlockMatcher(matcher_ldg).match(cfg=cfg)

matcher_hmmx2 = instructions_contain(instruction_is(
    Fp16MinMaxMatcher(pmax=True))
    .with_operand(index=1, operand=f'{matched_ldg[0].operands[0]}.H0_H0')
    .with_operand(index=2, operand=f'{matched_ldg[1].operands[0]}.H0_H0')
    .with_operand(index=0, operand=RegisterMatcher(special=False))
)
matched_hmmx2 = matcher_hmmx2.match(blk.instructions[matcher_ldg.next_index:])
```

Table 1: Comparison of the SASS code generated for the sm_100 architecture for the 16-bit `__half` (left) vs 32-bit float (right) maximum function.

<code>__hmax(const __half, const __half)</code>	<code>fmax(const float, const float)</code>
<code>LDG.E.U16 R2, desc[UR6][R2.64]</code>	<code>LDG.E.U16 R2, desc[UR6][R2.64]</code>
<code>LDG.E.U16 R5, desc[UR6][R4.64]</code>	<code>LDG.E.U16 R4, desc[UR6][R4.64]</code>
<code>...</code>	<code>...</code>
<code>HMNMX2 R5, R2.H0_H0, R5.H0_H0, !PT</code>	<code>HADD2.F32 R6, -RZ, R2.H0_H0</code>
<code>...</code>	<code>HADD2.F32 R7, -RZ, R4.H0_H0</code>
	<code>FMNMX R6, R6, R7, !PT</code>
	<code>F2FP.F16.F32.PACK_AB R3, RZ, R6</code>
	<code>...</code>
<code>STG.E.U16 desc[UR6][R6.64], R5</code>	<code>STG.E.U16 desc[UR6][R6.64], R3</code>

114 The binary analysis targeting SASS rather than PTX is a trade-off. PTX is fully documented
 115 ([NVIDIA Corporation, 2025f](#)) and architecture-independent, but it is an intermediate
 116 representation. SASS is the machine code the hardware executes; assertions on SASS thus
 117 characterize the final instructions and their scheduling. Inspecting SASS is also relevant for
 118 kernel profiling: performance metrics are directly associated with executed instructions, whereas
 119 interpreting them at a higher level requires correlation, such as through compiler-emitted
 120 line information. However, SASS is only partially documented and architecture-specific; the
 121 architecture-aware matcher hierarchy addresses architecture differences.

122 Research impact statement

123 Three getting-started examples in the documentation demonstrate the components end to end.
 124 The API tracing example programmatically verifies the sequence of runtime calls that realize a
 125 diamond-shaped dependency among four asynchronous kernel launches. The kernel profiling
 126 example analyzes the performance effect of a restructuring of a fill kernel and encodes the
 127 findings in test assertions. The binary analysis example asserts the SASS instructions to which
 128 an atomic function call compiles, with a single matcher specification that adapts across CUDA
 129 architectures. Run in CI/CD pipelines, such assertions can detect when an analysis ceases to
 130 hold as toolchains and libraries evolve.

131 ReProspect has been successfully used as a support for contributions to the open-source
 132 Kokkos library ([Trott et al., 2022](#)) and the development of an in-house finite element code
 133 built on top of the open-source Trilinos library ([Mayr et al., 2026](#)) ([Arnst & Tomasetti, 2024](#))
 134 ([Tomasetti & Arnst, 2024](#)).

135 The [examples directory](#) contains several case studies inspired by these research efforts.

136 Kokkos::View allocation

137 CUDA API tracing provides insight into microbenchmarking results assessing the behavior of
 138 Kokkos::View allocation.

139 See [online example](#) and [Kokkos issue](#).

140 Impact of Kokkos::complex alignment

141 Our in-house finite element code uses complex arithmetic for frequency-domain
 142 electromagnetism simulations. This example combines kernel profiling and SASS
 143 analysis to assess how aligning Kokkos::complex<double> to 8 or 16 bytes impacts memory
 144 instructions and traffic.

145 See [online example](#).

Dynamic dispatch for virtual functions on device

This example uses ReProspect to create a research artifact that reproduces the dynamic dispatch instruction pattern identified by (Zhang et al., 2021).

See [online example](#).

Atomics with desu1

Kokkos provides extended atomic support through the desu1 library (Trott et al., 2022), which maps atomic operations to one of several methods with varying performance. The choice of the method depends on intricate logic, and must be tested. A micro-benchmarking approach is feasible, but requires the physical device and suffers from runtime variability. Yet, the machine code already contains information about the selected code paths. This case study demonstrates how to verify which method is chosen by matching an instruction sequence pattern.

See [online example](#).

Code availability

ReProspect is available under the LGPL-3.0 license on [GitHub](#) and released on [PyPI](#). The documentation provides detailed instructions for building and running the tests and examples.

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AI usage disclosure

No AI was used for the design of the code. Alongside traditional tools such as `pylint` and `mypy`, Claude and CoPilot were used as assistants to improve implementation details of individual functions. AI helped improve the clarity of the manuscript and the documentation.

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